Mathieu Desroches

Curriculum Vitæ

September 2026

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1 Contact Details

1.1 Personal Information

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French Nationality

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1.2 Current Appointment

I am an **Research Director** (*Directeur de Recherche de Classe 2*, DR2) at Inria, in the Inria Branch at the University of Montpellier (France).
Since January 2016, I have been the **Head** of the MathNeuro project-team since January 2016.
Since January 2026, I have been **Head of Science** of the Inria Branch at the University of Montpellier.

1.3 Previous Appointments

Inria Sophia Antipolis - Méditerranée Research Centre, Valbonne, France.

Associate Research Professor (*Chargé de Recherche de Classe Normale*, CRCN),
MathNeuro project-team **January 2019 - June 2023**.

Associate Research Professor (*Chargé de Recherche de Classe 1*, CR1),
MathNeuro project-team **January 2016 - December 2018**.

Associate Research Professor (*Chargé de Recherche de Classe 1*, CR1),
NeuroMathComp project-team **October - December 2015**.

Inria Paris-Rocquencourt Research Centre, Rocquencourt, France.

Associate Research Professor (*Chargé de Recherche de Classe 1*, CR1),
Mycenae project-team **January 2014 - September 2015**.

Associate Research Professor (*Chargé de Recherche de Classe 1*, CR1),
Sisyphe project-team **December 2013**.

Assistant Research Professor (*Chargé de Recherche de Classe 2*, CR2),
Sisyphe project-team **November 2011 - November 2013**.

University of Bristol, Department of Engineering Mathematics, Bristol, UK.

Postdoctoral Fellow **April 2009 - October 2011**.

A. Chérioux Highschool, Vitry-sur-Seine, France.

Mathematics Teacher **Sept 2004 - April 2006**.

2 Education

2.1 Higher Education

University of Bristol, Department of Engineering Mathematics, Bristol, UK.

PhD. in Applied Mathematics **April 2006 - February 2009**.

- Thesis : "Numerical Continuation Methods for Slow-Fast Dynamical Systems"
- Supervisors : Prof. Hinke M. Osinga and Prof. Bernd Krauskopf, Bristol Centre for Applied Nonlinear Mathematics, Department of Engineering Mathematics, University of Bristol, Bristol, UK. *Now both at the University of Auckland, New Zealand*

Université D. Diderot, Équipe d'Analyse Algébrique, Paris, France.

DEA (Masters Degree) in Pure Mathematics **Sept 2004**.

- Graduated with honours.
- Course work in Riemannian Geometry, Symplectic Geometry, Dynamical Systems and Hamiltonian Dynamics.
- Thesis : "The Birkhoff-Smale Theorem ; Examples in Hamiltonian Dynamics"
- Supervisor : Professor Jacques Féjoz, Institut de Mathématiques de Jussieu (UMR 7586 du CNRS), équipe d'Analyse Algébrique, University P. & M. Curie, Paris, France. *Now at Université Paris-Dauphine, Paris, France.*

Université P. & M. Curie, Équipe d'Analyse Fonctionnelle, Paris, France.

External 'Agrégation' (competitive examination) in Mathematics **June 2003**.

DEA (Masters Degree) in Functional Analysis **June 2001**.

- Course work in Descriptive Set Theory and Geometry of Banach Spaces.
- Thesis : "A study of the paper *Smooth norms on certain* $C(K)$ *spaces* by Petr Hájek (*Proc. Amer. Math. Soc.* **131**(7): 2049–2051, 2003)"
- Supervisor : Professor Gilles Godefroy, Institut de Mathématiques de Jussieu (UMR 7586 du CNRS), équipe d'Analyse Fonctionnelle, University P. & M. Curie, Paris, France.

Maîtrise (fourth year diploma) in Pure Mathematics **June 2000**.

- Graduated with first class honours.
- Course work in Lie Groups and Algebras, Real and Applied Analysis, PDEs, Logic and Model Theory.

Licence (third year diploma) in Applied Mathematics **June 1999**.

- Graduated with first class honours.
- Course work in Matrix Analysis, Measure Theory, ODEs, Topology and Functional Analysis, Calculus and Complex Analysis.

A. Schweitzer Highschool, Le Raincy, France.

Classes Préparatoires Scientifiques **1996-1998**.
(Scientific Preparatory Classes)

A. Boulloche Highschool, Livry-Gargan, France.

Baccalauréat (série Scientifique, option Mathématiques) **June 1996**.

- graduated with Honours

2.2 Training Courses

Centre de Recherches Mathématiques, Université de Montréal, Canada.

Minicourse on Advanced Algorithms and Numerical Software for the Bifurcation Analysis of Dynamical Systems **30 June - 1 July, 2007**.

Department of Applied Mathematics, University of Twente, The Netherlands.

Multiscale Modeling and Singular Perturbations **7-11 Jan 2008**.

3 Research

3.1 Ph.D. (defended: 19 February 2009, awarded: 6 May 2009)

My PhD research dealt with the numerical study of global manifolds in slow-fast dynamical systems. This includes numerical computations of slow manifolds and canard orbits, bifurcation analysis of singularly perturbed systems, with a view towards applications, e.g., in biology, chemistry, or neuroscience.

- **Thesis :** Numerical Continuation Methods for Slow-Fast Dynamical Systems.
- **Supervisors :** Prof. Hinke M. Osinga and Prof. Bernd Krauskopf
  (University of Bristol, Bristol, UK, now at the University of Auckand, New Zealand).
- **Examiners :** Prof. Alan R. Champneys (University of Bristol, Bristol, UK)
  Prof. Tasso J. Kaper (Boston University, Boston, USA)

3.2 Habilitation thesis (defended: 11 December 2015, awarded: 10 february 2016)

- **Thesis :** Complex Oscillations with Multiple Time Scales. Application to Neuronal Dynamics.
- **Reviewers :** Prof. Eusebius J. Doedel (Concordia University, Montréal, Canada)
  Prof. Christopher K. R. T. Jones (University of North Carolina at Chapel Hill, USA)
  Prof. Daniel Panazzolo (Université de Haute-Alsace, France)
- **Jury :** Prof. Stephen Coombes (University of Nottingham, UK)
  Prof. Peter De Maesschalck (Hasselt University, Belgium)
  Prof. Olivier Faugeras (Inria Sophia Antipolis - Méditerranée, France)
  Prof. Jean-Pierre Françoise (Université Pierre et Marie Curie, France)
  Prof. Christopher K. R. T. Jones (University of North Carolina at Chapel Hill, USA)
  Prof. Daniel Pannazolo (Université de Haute-Alsace, France)

3.3 Research Interests

My research is focused on geometric singular perturbation theory and the canard phenomenon in the context of both smooth and non-smooth (essentially piecewise-linear, which in my opinion is the minimal setup) dynamical systems with a clear focus towards biological applications, in particular in Neuroscience where I am looking at the robustness of slow-fast phenomena in general, and canard behaviours in particular. Since my PhD work, I have been interested in complex oscillations with multiple timescales (bursting oscillations, mixed-mode oscillations, mixes of the two) and their patterns in parameter space, often organised by canard solutions. Within Mathematical Neuroscience, I design models in close collaboration with experimentalists, to address questions related to excitability and synaptic plasticity. In recent years, I have been developing multiscale models of neurotransmitter release and also of cortical spreading depression. In terms of computational techniques, I am interested in further developing numerical continuation methods adapted to the study of multiple timescale dynamics and to the computation of challenging dynamical objects (invariant manifolds, isolas, complex i.e. multi-mode and bursting oscillations). Finally, I am also looking at models where slow-fast dynamics appears without necessarily an explicit timescale separation, which leads me to work on theoretical and computational methods to study such systems, namely, ε-free systems; in particular, I have revisited inflection methods and their link to excitable dynamics.

3.4 Research Visits

Center for Applied Mathematics, Cornell University, USA.
(invited by Prof. John Guckenheimer) **January - March, 2009.**
27 October - 7 November, 2009.

Centre de Recerca Matemàtica, Barcelona, Spain.
(invited by Prof. Antoni Guillamon) **3-15 May, 2009.**

Laboratoire J.-L. Lions, Université P. & M. Curie, France.
(invited by Prof. Jean-Pierre Françoise) **11-18 Feb, 2010.**

INRIA, Centre de Paris - Rocquencourt, Équipe SISYPHE-REGATE, France.
(invited by Prof. Frédérique Clément) **4-11 May, 2010.**

Center for Biodynamics, Boston University, USA.
(invited by Prof. Tasso J. Kaper) **3-10 Oct, 2010.**

Department of Mathematics, Hasselt University, Belgium.
(invited by Prof. Peter De Maesschalck) **12-18 Dec, 2010.**

Center for Biodynamics, Boston University, USA.
(invited by Prof. Tasso J. Kaper) **30 May - 3 June, 2011.**

Department of Computer Science and Artificial Intelligence, University of Granada, Spain.
(invited by Prof. Jesus M. Cortes) **27 June - 1 July, 2011.**

Department of Mathematics, Technical University of Denmark, Denmark.
(invited by Prof. Morten Brøns) **15-20 August, 2011.**

Center for Biodynamics, Boston University, USA.
(invited by Prof. Tasso J. Kaper) **12 - 18 May, 2013.**

Department of Mathematics, University of the Balears Islands, Palma de Mallorca, Spain.
(invited by Prof. Antonio Teruel) **22 - 18 July, 2013.**

Department of Mathematics, University of the Balearic Islands, Palma de Mallorca, Spain.
(invited by Prof. Antonio Teruel) **1 - 27 June, 2014.**

Department of Mathematics, Technical University of Denmark, Denmark.
(invited by Prof. Morten Brøns) **20-26 November, 2015.**

Department of Mathematics, University of the Balearic Islands, Palma de Mallorca, Spain.
(invited by Prof. Antonio Teruel) **15 June - 13 July, 2016.**

Department of Mathematics, University of Auckland, Auckland, New-Zealand.
(invited by Prof. Vivien Kirk) **22 August - 4 September, 2016.**

Basque Centre for Applied Mathematics, Bilbao, Spain.
(invited by Prof. Serafim Rodrigues) **08 June - 06 July, 2017.**
28 August - 12 September, 2017.
18 June - 18 July, 2018.
10 July - 10 August, 2019.
28 June - 18 July and 01-20 August, 2020.
05 Sept. - 05 Oct., 2022.
01 July - 31 Aug., 2023.
15 July - 15 Aug., 2024.
07 July - 20 July, 2025.

4 Publications

3437 citations and an h-index of 31 as of September 2026 [source: Google Scholar]

4.1 Preprints (in revision, submitted or to be submitted)

[Pr1] B. Kula, **M. Desroches**, S. Rodrigues and M. Kukley
Geometric principles determining the morphology of oligodendrocyte precursor cells in brain white matter, In revision in Glia, 2026.

[Pr2] D. Barros de Souza, J. Teodomiro, F.A.N. Santos, M. Ding, W. Sun, **M. Desroches**, J. Jöst and S. Rodrigues, *Efficient decomposition of Forman-Ricci curvature over Vietoris-Rips complexes and Data Applications*, In revision in SIAM J. Mathematics of Data Science, 2026.

[Pr3] J. Areias, Y. Chastagnier, J. Naudé, **M. Desroches**, J. Perroy and V. Szabo
Sevoflurane anesthesia triggers immediate-early gene expression and low-threshold spike bursting in the mouse paraventricular thalamus, Submitted, 2026.

[Pr4] R. Cavalcanti, N. L. Santos Junior, F. A. N. Santos, **M. Desroches** and S. Rodrigues
Topological Data Analysis in Finsler Spaces, Submitted, 2026.

4.2 Book chapters

[Ch2] T. Fülöp, C. Ramassamy, G. Lacombe, E. H. Frost, A. A. Cohen, S. Rodrigues, **M. Desroches**, K. Hirokawa, B. Laurent and J. M. Witkowski, *Infection, Neuroinflammation and Interventions for Healthy Brain and Longevity*, In: Brain and Mental Health in Ageing, G. Kaur and S. I. S. Rattan, Eds., Springer, 2024, pp. 255–275.

[Ch1] **M. Desroches**, S. Fernández-García, M. Krupa, R. Prohens and A. E. Teruel
Piecewise-linear (PWL) canard dynamics: Simplifying singular perturbation theory in the canard regime using piecewise-linear systems, In: Nonlinear Systems, Vol. 1: Mathematical Theory and Computational Methods, V. Carmona, J. Cuevas-Maraver, F. Fernández-Sánchez and E. García-Medina, Eds., Springer, 2018, pp. 67–86.

4.3 Peer-reviewed journal articles

[84] H. Soydemir, C. Gauld, **M. Desroches**, B. Pignon and D. Depannemaecker
Dynamical System Modeling in Schizophrenia: A Narrative Review of Computational Psychiatry Frameworks. Dialogues in Clinical Neuroscience (accepted), 2026.

[83] E. Baspinar, D. Avitabile, F. Campillo, C. Nouveau, **M. Desroches** and M. Mantegazza
An astro-neural-field model for cortical spreading depression.
Journal of Computational Neuroscience (accepted), 2026.

[82] D. Fedorov, F. Campillo, **M. Desroches**, E. Soria-Gómez, J. Piriz and S. Rodrigues
Multiple bursting patterns in Lateral Habenula neurons: experiments and computational model
The Journal of Physiology **604**(8): 3396-3412, 2026.

[81] A. A. Cohen, E. Frost, A. Khalil, S. Rodrigues, **M. Desroches**, M. Alami, K. Hirokawa, J. Witkowski, C. Ramassamy and T. Fülöp, *Unmasking the hidden catalyst: how infections trigger Alzheimer's Disease*
Journal of Alzheimer's Disease **110**(2): 519-536, 2026.

[80] M. Ding, J. Yu, D. Souza, S. Rodrigues, **M. Desroches**, C. Li and W. Sun
Growing order-heterogeneous simplicial complexes, Chaos **36**(1): 013143, 2026.

[79] J. Danielewicz, A. V. Chizhov, G. Girier, **M. Desroches**, J. M. Encinas and S. Rodrigues
Fluid excitability and "flipping" of granule cells: An experimental and computational study
PLOS One **21**(2): e0339418, 2026.

[78] D. V. Amakhin, A. V. Chizhov, G. Girier, **M. Desroches**, J. Sieber and S. Rodrigues
Observing hidden neuronal states in experiments
PLOS Computational Biology **21**(12): e1013748, 2025.

[77] L. Lemaire, **M. Desroches**, S. Rodrigues and F. Campillo
Depolarization block induction via slow $Na_V1.1$ inactivation in Dravet syndrome
Scientific Reports **15**(1): 32749, 2025.

[76] E. Baspinar, M. Simonti, H. Srour, **M. Desroches**, D. Avitabile and M. Mantegazza
GABAergic neurons can facilitate the propagation of cortical spreading depolarization: experiments in mouse neocortical slices and a novel neural field computational model
PLOS Computational Biology **21**(6): e1013099, 2025.

[75] H. Todd, **M. Desroches**, A. Cayco-Gajic and B. Gutkin
The role of gap junctions and clustered connectivity in emergent synchronisation patterns of inhibitory neuronal networks, Physical Review E **112**: 014405, 2025.

[74] D. Depannemaecker, F. Tesler, **M. Desroches**, V. Jirsa and A. Destexhe
Modeling impairment of ionic regulation with extended Adaptive Exponential integrate-and-fire models
Journal of Computational Neuroscience **53**(1): 1-8, 2025.

[73] G. Rabuffo, A. Bandyopadhyay, C. Calabrese, K. Gudibanda, D. Depannemaecker, L. Mitiko Takarabe, M.-L. Saggio, **M. Desroches**, A. Ivanov, M.-L . Linne, C. Bernard, S. Petkoski and V. K. Jirsa
Biophysically inspired mean-field model of neuronal populations driven by ion exchange mechanisms
eLife **14**, 2025.

[72] **M. Desroches**, P. Kowalczyk and S. Rodrigues
Discontinuity-induced dynamics in the Conductance-Based Adaptive Exponential Integrate-and-Fire Model
Bulletin of Mathematical Biology **87**(1): 2, 2025.

[71] J. Penalva, **M. Desroches**, A. E. Teruel and C. Vich
Dynamics of a piecewise-linear Morris-Lecar model: bifurcations and spike-adding phenomena
Journal of Nonlinear Science **34**(3): 52, 2024.

[70] E. Pavlidis, F. Campillo, A. Goldbeter and **M. Desroches**
Multiple-timescale dynamics, mixed mode oscillations and mixed affective states in a model of Bipolar Disorder, Cognitive Neurodynamics **18**(6): 3239-3257, 2024.

[69] T. Fülöp, (...), **M. Desroches**, S. Rodrigues et al.
Viruses - a major cause of amyloid deposition in the brain
Expert Review of Neurotherapeutics **23**(9): 775-790, 2023.

[68] A. V. Chizhov, D. V. Amakhin, A. Erdem Sagtekin, **M. Desroches**
Single-compartment model of a pyramidal neuron, fitted to recordings with current and conductance injection, Biological Cybernetics **117**: 433-451, 2023.

[67] M. Yamakou, **M. Desroches** and S. Rodrigues
Synchronization in STDP-driven memristive neural networks with time-varying topology
Journal of Biological Physics **49**: 483-507, 2023.

[66] M. Sensi, **M. Desroches** and S. Rodrigues
Slow-fast dynamics in a neurotransmitter release model: delayed response to a time-dependent input signal
Physica D **455**: 133887, 2023.

[65] G. Girier, **M. Desroches** and S. Rodrigues
From integrator to resonator neurons: A new multiple-timescale scenario
Nonlinear Dynamics **111**: 16545-16556, 2023.

[64] L. Lemaire **M. Desroches**, M. Krupa and M. Mantegazza
Idealized multiple-timescale model of CSD initiation and pre-epileptic hyperexcitability caused by NaV1.1/SCN1A mutations, Journal of Mathematical Biology **86**(1): 92, 2023.

[63] J. Penalva, **M. Desroches**, A. E. Teruel and C. Vich
Slow passage through a Hopf-like bifurcation in piecewise linear systems: application to elliptic bursting.
Chaos **32**(12): 123109, 2022.

[62] A. Guidolin, **M. Desroches**, J. D. Victor, K. P. Purpura and S. Rodrigues,
Geometry of spiking patterns in early visual cortex: a Topological Data Analytic approach.
Journal of the Royal Society Interface **19**(196): 20220677, 2022.

[61] D. Avitabile, **M. Desroches** and G. B. Ermentrout
Cross-scale excitability in networks of quadratic integrate-and-fire neurons
PLOS Computational Biology **18**(10): e1010569, 2022.

[60] K. Bourgade, (...), **M. Desroches**, S. Rodrigues and T. Fülöp
Interaction Mechanism Between the HSV-1 Glycoprotein B and the Antimicrobial Peptide Amyloid-β
Journal of Alzheimer's Disease Reports **6**(1): 599-606, 2022.

[59] H. Taher, D. Avitabile and **M. Desroches**
Bursting in a next generation neural mass model with synaptic dynamics: a slow-fast approach
Nonlinear Dynamics **108**: 4261–4285, 2022.

[58] A. Earls, M.-C. Calderer, **M. Desroches**, A. Zarnescu and S. Rodrigues
A phenomenological model for interfacial water near hydrophilic polymers
Journal of Physics: Condensed Matter **34**: 355102, 2022.

[57] T. Fülöp, (...), S. Rodrigues, **M. Desroches** *et al.*
Immunosenescence and Altered Vaccine Efficiency in Older Subjects: A Myth Difficult to Change,
Vaccines **10**(4): 10040607, 2022.

[56] **M. Desroches**, J. Rinzel and S. Rodrigues
Classification of bursting patterns: A tale of two ducks
PLOS Computational Biology **18**(2): e1009752, 2022.

[55] O. Chever, S. Zerimech, P. Scalmani, L. Lemaire, A. Loucif, M. Ayrault, M. Krupa,
M. Desroches, F. Duprat, S. Cestèle and M. Mantegazza
GABAergic neurons and $\mathrm{Na_V}1.1$ *channel hyperactivity: a novel neocortex-specific mechanism of Cortical Spreading Depression*
Journal of Clinical Investigation **131**(21): e142203, 2021.

[54] L. Lemaire, **M. Desroches**, M. Krupa, L. Pizzamiglio, P. Scalmani and M. Mantegazza
Modeling $\mathrm{Na_V}1.1$/SCN1A *sodium channel mutations in a microcircuit with realistic ion concentration dynamics suggests differential GABAergic mechanisms in epilepsy and migraine*
PLOS Computational Biology **17**(7): e1009239, 2021.

[53] U. Munawara, (...), **M. Desroches**, S. Rodrigues and T. Fülöp
Hyperactivation of monocytes and macrophages in MCI patients contributes to the progression of Alzheimer's Disease, Immunity and Ageing **18**: 29, 2021.

[52] T. Fülöp, S. Tripathi, S. Rodrigues, **M. Desroches** *et al.*
Targeting Impaired Antimicrobial Immunity in the Brain for the Treatment of Alzheimer's Disease
Neuropsychiatric Disease and Treatment **17**: 1311–1339, 2021.

[51] E. Baspinar, D. Avitabile and **M. Desroches**
Canonical models for torus canards in elliptic bursters
Chaos **31**(6): 063129, 2021.

[50] **M. Desroches**, P. Kowalckyk and S. Rodrigues
Spike-adding and reset-induced canard cycles in adaptive integrate and fire models
Nonlinear Dynamics **104**(3): 2451–2470, 2021.

[49] T. Fülöp, **M. Desroches**, A. A. Cohen, F. A. N. Santos and S. Rodrigues
Why we should use topological data analysis in ageing: Towards defining the "topological shape of ageing".
Mechanisms of Ageing and Development **192**: 111390, 2020.

[48] J. Uria-Albizuri, **M. Desroches**, M. Krupa and S. Rodrigues
Inflection, canards and folded singularities lines in excitable systems – Application to a 3D FitzHugh-Nagumo model. Journal of Nonlinear Science **30**(6): 3265–3291, 2020.

[47] T. Fülöp, **M. Desroches**, S. Rodrigues *et al.*
Targeting Infectious Agents as a Therapeutic Strategy in Alzheimer's Disease.
CNS Drugs **34**(7): 673–695, 2020.

[46] D. Avitabile, **M. Desroches**, R. Veltz and M. Wechselberger
General local theory for spatiotemporal canards and delayed bifurcations
SIAM Journal on Mathematical Analysis **52**(6): 5703–5747, 2020.

[45] R. Barrio, S. Coombes, **M. Desroches**, F. Fenton, S. Luther and E. Pueyo
Excitable dynamics in neural and cardiac systems (Introduction to special issue)
Communications in Nonlinear Science and Numerical Simulation **86**: 105275, 2020.

[44] E. Köksal Ersöz, **M. Desroches**, A. Guillamon, J. Rinzel and J. Tabak
Canard-induced complex oscillations in an excitatory network
Journal of Mathematical Biology **80**(7): 2075–2107, 2020.

[43] **M. Desroches**, O. Faugeras, M. Krupa and M. Mantegazza
Modeling cortical spreading depression induced by the hyperactivity of interneurons.
Journal of Computational Neuroscience **47**(2-3): 125–140, 2019.

[42] P. beim Graben, A. Jimenez-Marin, I. Diez, J. M. Cortes, **M. Desroches** and S. Rodrigues
Metastable resting state brain dynamics
Frontiers in Computational Neuroscience **13**: 62, 2019.

[41] A. Chizhov, F. Campillo, **M. Desroches**, A. Guillamon and S. Rodrigues
Conductance-based refractory density approach for a population of bursting neurons
Bulletin of Mathematical Biology **81**(10): 4124–4143, 2019.

[40] **M. Desroches**, J.-P. Françoise and M. Krupa
Parabolic bursting, spike-adding, dips and slices in a minimal model
Mathematical Modelling of Natural Phenomena **14**(4), 2019.

[39] E. Köksal Ersöz, **M. Desroches**, C. Mirasso and S. Rodrigues
Anticipation via canards in excitable systems
Chaos **29**(1): 013111, 2019.

[38] **M. Desroches** and V. Kirk
Spike-adding in a canonical three-timescale system: superslow explosion and folded-saddle canards
SIAM Journal on Applied Dynamical Systems **17**(3): 1989–2017, 2018.

[37] T. Lissek *et al.*
Building Bridges through Science
Neuron **96**(4): 730–735, 2017. [Opinion Letter]

[36] D. Avitabile, **M. Desroches** and E. Knobloch
Spatio-temporal canards in neural field equations
Physical Review E **95**(4): 042205, 2017.

[35] D. Avitabile, **M. Desroches**, E. Knobloch and M. Krupa
Ducks in space : from nonlinear absolute instability to noise-sustained structures in a pattern-forming system. Proceedings of the Royal Society A **473**(2207): 20170018, 2017.

[34] G. Carmantini, P. Beim Graben, **M. Desroches** and S. Rodrigues
A modular architecture for transparent computation in recurrent neural networks
Neural Networks **85**(1): 85–105, 2017.

[33] E. Köksal-Ersöz, **M. Desroches** and M. Krupa
Synchronization of weakly coupled canard oscillators
Physica D: Nonlinear Phenomena **349**: 46–61, 2017.

[32] **M. Desroches**, A. Guillamon, E. Ponce, R. Prohens, S. Rodrigues and A. E. Teruel
Canards, folded nodes and mixed-mode oscillations in piecewise-linear slow-fast systems
SIAM Review **58**(4): 653–691, 2016.

[31] S. Rodrigues*, **M. Desroches***, M. Krupa*, J. M. Cortes, T. J. Sejnowski and A. B. Ali*
Time-coded neurotransmitter release at excitatory and inhibitory synapses (* joint first author)
Proceedings of the National Academy of Sciences of the USA **113**(8): E1108–E1115, 2016.

[30] **M. Desroches**, M. Krupa and S. Rodrigues
Spike-adding in parabolic bursters: the role of folded-saddle canards
Physica D: Nonlinear Phenomena **331**: 58–70, 2016.

[29] **M. Desroches**, O. Faugeras and M. Krupa
Slow-fast transitions to seizure states in the Wendling-Chauvel neural mass model
Opera Medica et Physiologica **2**(3): 228–234, 2016.

[28] **M. Desroches**, F. Fernández-García and M. Krupa
Canards and spike-adding in a minimal piecewise-linear square-wave burster
Chaos **26**(7): 073111, 2016.

[27] E. Köksal Ersöz, **M. Desroches**, M. Krupa and F. Clément
Canards-mediated (de)synchronization in coupled phantom bursters
SIAM Journal on Applied Dynamical Systems **15**(1): 580–608, 2016.

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From canards of folded singularities to torus canards in a forced van der Pol equation
Journal of Nonlinear Science **26**(2): 405–451, 2016.

[25] F. Fernández-García, **M. Desroches**, M. Krupa and A. E. Teruel
Canards in planar piecewise-linear slow-fast systems with three zones
Dynamical Systems: an International Journal **31**(2): 173–197, 2015.

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Extending the zero-derivative principle for slow-fast dynamical systems
Zeitschrift für angewandte Mathematik und Physik (ZAMP) **66**(5): 2255–2270, 2015.

[23] F. Fernández-García, **M. Desroches**, M. Krupa and F. Clément
A slow-fast coupling of planar piecewise-linear oscillators; Application to GnRH secretion model
SIAM Journal on Applied Dynamical Systems **14**(2): 643–673, 2015.

[22] J. Touboul, M. Krupa and **M. Desroches**
Noise-induced canard and mixed-mode oscillations in large-scale stochastic networks
SIAM Journal on Applied Mathematics **75**(5): 2024–2049, 2015.

[21] M. Brøns, **M. Desroches** and M. Krupa
Mixed-Mode Oscillations due to a singular Hopf bifurcation in a forest pest model
Mathematical Population Studies **22**(1): 71–79, 2015.

[20] I. M. De la Fuente, J. M. Cortes, E. Valero, **M. Desroches**, S. Rodrigues, I. Malaina and L. Martinez
On the dynamics of the Adenylate Energy System: Homeorhesis vs Homeostasis.
PLOS One **9**(10): e108676, 2014.

[19] **M. Desroches**, M. Krupa and T. J. Kaper
Mixed-Mode Bursting Oscillations: dynamics created by a slow-passage through spike-adding canard explosion in a square-wave burster
Chaos **23**(4): 046106, 2013.

[18] J. M. Cortes*, **M. Desroches***, S. Rodrigues*, R. Veltz, M. A. Muñoz and T. J. Sejnowski
Short-term synaptic plasticity in the deterministic Tsodyks-Markram model leads to unpredictable network dynamics (* joint first author)
Proceedings of the National Academy of Science of the USA **110**(41): 16610–16615, 2013.

[17] P. De Maesschalck and **M. Desroches**
Numerical continuation techniques for planar slow-fast systems
SIAM Journal on Applied Dynamical Systems **12**(3): 1159–1180, 2013.

[16] **M. Desroches**, E. Freire, S. J. Hogan, E. Ponce and P. Thota
Canards in piecewise-linear systems: explosions and super-explosions
Proceedings of the Royal Society A **469**(2154): 20120603, 2013.

[15] J. Sieber, A. Rapaport, S. Rodrigues and **M. Desroches**
A new method for the reconstruction of unknown non-monotonic growth functions in the chemostat
Bioprocess and Biosystems Engineering **36**(10): 1497–1507, 2013.

[14] **M. Desroches**, M. Krupa and S. Rodrigues
Canards, inflection and excitability threshold in neuronal models
Journal of Mathematical Biology **67**(4): 989–1017, 2013.

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Mixed-mode oscillations in a multiple time scale phantom bursting system
SIAM Journal on Applied Dynamical Systems **11**(4): 1458–1498, 2012.

[12] D. Avitabile, **M. Desroches**, and S. Rodrigues
On the numerical continuation of isolas of equilibria
International Journal of Bifurcation and Chaos **22**(11): 1250277, 2012.

[11] D. Linaro, A. R. Champneys, **M. Desroches** and M. Storace
Spike-adding and codimension-two bifurcations in the Hindmarsh-Rose burster
SIAM Journal on Applied Dynamical Systems **11**(3): 939–962, 2012.

[10] **M. Desroches**, J. Guckenheimer, B. Krauskopf, C. Kuehn, H. M. Osinga and M. Wechselberger
Mixed-mode oscillations with multiple time scales (preprint)
SIAM Review **52**(2): 211–288, 2012.

[09] **M. Desroches**, J. Burke, T. J. Kaper and M. A. Kramer
Canards of mixed type in a neural burster
Physical Review E **85**(2): 021920, 2012.

[08] J. Burke, **M. Desroches**, A. M. Barry, T. J. Kaper and M. A. Kramer
A showcase of torus canards in neuronal bursters
The Journal of Mathematical Neuroscience **2**: 3, 2012.

[07] **M. Desroches** and M. R. Jeffrey
Canard and curvature : the "smallness of ε" in slow-fast dynamics (preprint)
Proceedings of the Royal Society A **467**(2132): 2404, 2011.

[06] **M. Desroches** and M. R. Jeffrey
Canard and curvature : nonsmooth approximation by pinching (preprint)
Nonlinearity **24**(5): 1655-1682, 2011.

[05] J. Rankin, **M. Desroches**, B. Krauskopf and M. Lowenberg
Canard cycles in aircraft ground dynamics (preprint)
Nonlinear Dynamics **66**(4): 681–688, 2011.

[04] **M. Desroches**, B. Krauskopf and H. M. Osinga
Numerical continuation of canard orbits in slow-fast dynamical systems (preprint)
Nonlinearity **23**(3): 739–765, 2010.

[03] **M. Desroches**, B. Krauskopf and H. M. Osinga
The geometry of mixed-mode oscillations in the Olsen model for P.-O. reaction (preprint)
Discrete and Continuous Dynamical Systems Series S **2**(4): 807–827, 2009.

[02] **M. Desroches**, B. Krauskopf and H. M. Osinga
The geometry of slow manifolds near a folded node (preprint)
SIAM Journal on Applied Dynamical Systems **7**(4): 1131–1162, 2008.

[01] **M. Desroches**, B. Krauskopf and H. M. Osinga
Mixed-mode oscillations and slow manifolds in the self-coupled FitzHugh-Nagumo system (preprint)
Chaos **18**(1): 015107, 2008.

4.4 Peer-reviewed conference proceedings papers

[Proc6] G. Carmantini, P. Beim Graben, **M. Desroches** and S. Rodrigues
Turing Computation with Recurrent Artificial Neural Networks.
NIPS workshop — Cognitive Computation: Integrating Neural and Symbolic Approaches (Montréal, 7-12 December 2015), 2016.

[Proc5] **M. Desroches**, J.-P. Françoise and L. Mégret
Canard-induced loss of stability across a homoclinic bifurcation
ARIMA, Proceedings of a conference in honour of Éric Benoît (La Rochelle, France, 16-18 Dec. 2013), 2015.

[Proc4] A. Rapaport, J. Sieber, S. Rodrigues and **M. Desroches**
Extremum seeking via continuation techniques for optimising biogas production in the chemostat
NOLCOS 2013, 9th IFAC Symposium on Nonlinear Control Systems, Toulouse (France), 2013.

[Proc3] A. Rapaport, J. Sieber, S. Rodrigues and **M. Desroches**
A new method for the reconstruction of unknown non-monotonic growth functions in the chemostat
Proceedings of the IEEE Mediterranean Conference on Control and Automation (MED), Barcelona (Spain), July 2012.

[Proc2] D. Avitabile, **M. Desroches** and S. Rodrigues
Numerical continuation of isolas of equilibria
7th European Nonlinear Oscillations Conference (ENOC), Rome (Italy), July 2011.

[Proc1] P. Thota and **M. Desroches**
Numerical continuation of canard-like orbits in piecewise-linear systems
7th European Nonlinear Oscillations Conference (ENOC), Rome (Italy), July 2011.

4.5 Work in Progress

- M. Meertens, **M. Desroches**, V. Kirk and C. Postlesthwaite
Folded homoclinic bifurcations as a route to spiking and bursting, In progress, 2026.
- **M. Desroches** and S. Rodrigues
Towards a new paradigm for brain-like sequential computations, In progress, 2026.

5 Talks

5.1 Invited Plenary talks

2016
- **9 June -** *Simplifying singular perturbation theory in the canard regime with piecewise-linear dynamics; application to neuronal models.* 10th NoLineal Conference, IMUS, Universidad de Sevilla (Spain).

5.2 Invited Lectures & Mini-courses

2018
- **18-22 June -** *Introduction to Mathematical Neuroscience: neuronal models and their bifurcations.* Course co-taught with Serafim Rodrigues, organised by the Basque Center for Applied Mathematics (BCAM) (Bilbao, Spain).

2013
- **9 December -** *Computation of slow manifolds and canards by numerical continuation of boundary-value problems.* Winter School organised by the Technical University of Denmark (Denmark).

2012
- **25 July -** *Mathematics for oscillations: introduction to numerical bifurcation analysis.* ICS Summer School 2012, organised by the Université Pierre et Marie Curie - Paris 6, Roscoff (France).
- **6-7 February -** *Canard-like orbits in PWL systems.* Stability and Bifurcations in Piecewise-Linear Systems, IMUS DOC-COURSE, Sevilla (Spain), 30 Jan.- 10 Feb., 2012.

2011
- **18 October -** *Introduction to numerical simulation and continuation with the software XPPAUT.* University of Granada (Spain).

5.3 Invited Presentations

2026
- **23 November -** *Modeling channelopathies with multiscale dynamical systems*, Invited seminar at the Physiology & Experimental Medicine of the Heart and Muscles, Montpellier (France).
- **2 September -** *Modelling multiple-timescale dynamics in life science & health: Slow, Fast and Sudden*, Invited talk at the IDIL Graduate Program Summer School 2026, Montpellier (France).
- **2 June -** *Observing hidden neuronal states in experiments*, Invited talk in the Minisymposium "Multiple Timescale Dynamics in Neuronal Systems (Part II)", at the International Conference on Mathematical NeuroScience (ICMNS), Montréal (Canada).
- **20 January -** *Classification of bursting oscillations via slow-fast dynamics*, Invited talk at the MACS Seminar, Department of Mathematics, University of Montpellier, Montpellier (France).

2025

- **09 October -** *Complex neuronal bursting oscillations: the role of slow variables*, Invited talk at the Coupled 80 Conference, Porto (Portugal).
- **18 September -** *Complex neuronal bursting oscillations: the role of slow variables*, Invited talk at the Seconde journées maths-bio de la fédération Occimath, Université Paul Sabatier, Toulouse (France).
- **13 June -** *Complex neuronal bursting oscillations: the role of slow variables*, Invited talk at the Virtual SMB MathNeuro Mini-Conference, Online.
- **12 May -** *Observing hidden neuronal states in experiments*, Invited talk in the minisymposium Patterns of Neural Activity - Part I of II, SIAM Conference on Application of Dynamical Systems, Denver (USA).
- **18 April -** *Three stories in multiscale modeling of neural systems*, Invited seminar at the Institute of Functional Genomics of Montpellier, Montpellier (France).
- **5 February -** *Classifying bursting oscillations using slow-fast dynamics*, Invited talk in the Applied Mathematics Webinar, LAMSIN (Tunisie).
- **24 January -** *Multiscale modeling of differential neurotransmitter release*, Invited talk at the 1st "Brain Plasticity & Modelisation" workshop, Montpellier (France).

2024

- **16 May -** *Complex neuronal bursting oscillations: the role of slow variables*, Invited talk in the Galvani in-person meeting, Saint Malo (France).
- **11 June -** *Observing hidden neuronal states in experiments*, Invited talk at the International Conference on Mathematical NeuroScience (ICMNS), Dublin (Ireland).
- **03 November -** *Endocannabinoids and neurodegeneration: two examples of collaboration between mathematicians and biologists*, Invited talk at the Mathematics and Life Health Forum, Shanghai (China).
- **04 November -** *Classification of complex oscillations in slow-fast dynamical systems with one and two slow variables*, Invited talk in the School of Mathematical Sciences, Shanghai Jiao Tong University, Shanghai (China).
- **22 November -** *Three stories in multiscale modeling of neural systems*, Invited seminar at the Institute of Neuroscience of Montpellier, Montpellier (France).

2023

- **21 September -** *Classification of Bursting Patterns: Review & extension*, Invited seminar talk at the Institut de Neuroscience des Systèmes, Marseille (France).
- **17 May -** *Classification of Bursting Patterns: Mind the slow variables*, Invited talk in the minisymposium Data Driven Modelling and Numerical Analysis of Complex Systems - Part III of III, SIAM Conference on Application of Dynamical Systems, Portland (USA).

2022

- **12 September -** *Classification of Bursting Patterns: A Tale of Two Ducks*, Invited online talk at the Mathematical Biology Seminar of the Mathematics Dept, University of Iowa, Iowa City, USA.
- **14 July -** *Classification of Bursting Patterns: A Tale of Two Ducks*, Invited talk in the Mini-Symposium on "Multiscale Oscillations in Biological Systems" at the Hybrid: SIAM Conference on the Life Sciences (LS22), Pittsburgh, USA.
- **28 June -** *Cross-scale excitability in networks of synaptically-coupled quadratic integrate-and-fire neurons*, Invited talk at the 40th congress of the francophone society for theoretical biology (SFBT), Poitiers, France.
- **3 February -** *Understanding Synaptic Mechanisms: Why a Multi-disciplinary Approach is Important?*, Invited joint talk with Afia B Ali (UCL, UK), at the Workshop on Mathematical Modeling and Statistical Analysis in Neuroscience, Institut Henri Poincaré, Paris, France. YouTube Link (starts at 4:40:00).
- **19 January -** *Slow-fast analysis of neural bursters: old & new*, Invited talk in the online seminar of the Société Française de Biologie Théorique (SFBT). YouTube Link.

2021

- **25 November -** *Canards in neural networks and their mean-field limits*, Invited talk in the Lowlands Seminar at VU Amsterdam, Amsterdam, The Netherlands.

2020

- **2 September -** *Slow-fast funnels - Towards analog brain-like computations*, Invited talk in the Neuromorphic Devices and Systems Group at IBM Research Zurich, Rüschlikon, Switzerland.
- **22 May -** *Slow-fast analysis of neural bursters: old & new*, Invited talk at the CAMBAM online seminar, McGill University, Montréal, Canada. YouTube link.

2019

- **10 November -** *Can the Topology-Structure-Function relationship help understand ageing?* - Invited talk in the International Symposium *Biology of Aging: Understanding aging to better intervene*, Montréal (Canada).
- **4 November -** *Novel mechanism of migraine-linked cortical spreading depression* (joint talk with M. Mantegazza) - Invited talk in the Workshop *IPMC: 30 years of biology on the riviera*, Sophia Antipolis (France).
- **5 September -** *Canards in excitatory networks.* - Invited talk in the mini-symposium Recent Advances in Slow-Fast Dynamics, XXXIXth Dynamics Days Europe Conference, Rostock (Germany).
- **19 July -** *Canards and spike-adding in neural bursters.* - Invited talk in the mini-symposium Mathematical Neuroscience, International Congress on Industrial and Applied Mathematics, Valencia (Spain).
- **22 May -** *Canards in excitatory networks.* - Invited talk in the mini-symposium Geometric Singular Perturbation Theory Beyond the Standard Form - Part I of II, SIAM Conference on Application of Dynamical Systems, Snowbird (USA).
- **22 February -** *Slow-fast analysis of bursting oscillations: old and new.* - Invited talk at the Bilateral International Meeting UK-France, Royal Society, Chicheley Hall, Milton Keynes (UK).

2018

- **29 May -** *Canards and spike-adding in neural bursters.* - Invited Keynote talk at the conference MURPHYS-HSFS-2018 meeting, Centre de Recerca Matemàtica, Barcelona (Spain).
- **6 June -** *Three-Timescale Dynamics: Canards and Spike Adding.* - Invited Keynote talk at the conference New Trends in Mathematical Biology, Centre de Recerca Matemàtica, Barcelona (Spain).

2017

- **22 May -** *Three-Timescale Dynamics: Canards and Spike Adding.* - Invited talk in the mini-symposium Recent Advances in Slow-Fast Dynamics, SIAM Conference on Application of Dynamical Systems, Snowbird (USA).
  Co-organiser of this 2-part mini-symposium with Daniele Avitabile and Vivien Kirk
- **29 March -** *Canards and spike-adding phenomena in neural bursters.* - Invited seminar talk, Applied Mathematics Seminar, The University of Liverpool, UK.
- **14 March -** *Canards and spike-adding phenomena in neural bursters.* - Invited seminar talk, Mathematical Neuroscience Seminar, University of the Balearic Islands, Palma, Spain.

2016

- **14 November -** *Understanding synaptic mechanisms: why a multi-disciplinary approach is important.* - Invited talk in the SFN Symposia meeting, New Techniques in Electro- and Optophysiology, San Diego, USA.
- **1 September -** *Canards in planar piecewise-linear slow-fast systems.* - Invited seminar talk, Applied Mathematics Seminar, University of Auckland, New Zealand.
- **24 August -** *Canards in piecewise-linear slow-fast systems.* - Invited seminar talk, Applied Mathematics Seminar, University of Sydney, Australia.
- **28 July -** *Spike-adding in parabolic bursters: the role of folded-saddle canards.* - Invited talk in the workshop "Dynamics in Life Sciences, Neuroscience", Volga Neuroscience meeting 2016, St-Petersburg / Nizhny-Novgorod (Russia).

- **14 July -** *Canards, folded singularities and bursting.* - Invited talk in the mini-symposium Slow-fast dynamics in neuroscience, 10th ECMTB Conference, Nottingham (UK).

2015

- **24 November -** *Piecewise-linear (PWL) slow-fast dynamical systems in the canard regime: Application to neuronal models.* - Invited seminar talk, Department of Mathematics and Computer Science, Technical University of Denmark, Copenhagen (Denmark).
- **12 June -** *Finding canard phenomena in the PieceWise-Linear world.* - Invited talk in the workshop Ecuaciones diferenciales y sistemas dinámicos: Una visión desde la Ingeniería II on the occasion of the 60th birthday of Enrique Ponce, Universidad de Sevilla, Sevilla (Spain).
- **21 May -** *Mixed-Mode Bursting Oscillations (MMBOs): Slow Passage Through a Spike-adding Canard Explosion.* - Invited talk in the mini-symposium Recent Advances on Multiple Timescale Dynamics, SIAM Conference on Application of Dynamical Systems, Snowbird (USA).
  Co-organiser of this 2-part mini-symposium with Vivien Kirk and Jonathan E. Rubin.
- **27 March -** *Piecewise-linear systems approximation of neuron models.* - Invited seminar talk, Séminaire Biocore, Inria Sophia Antipolis - Méditerranée Research Centre (France).
- **17 February -** *Mixed-Mode Bursting Oscillations (MMBOs): slow passage through spike-adding canard explosion in a square-wave burster.* - Invited talk in the workshop Quantitative Biomedicine for Health and Disease, BCAM, Bilbao (Spain).

2014

- **10 June -** *Canards in 3D piecewise-linear systems - Application to neuron models.* - Invited talk in the special session Nonlinear dynamics in neuroscience, The 10th AIMS Conference on Dynamical Systems, Differential Equations and Applications, Madrid (Spain).
- **7 March -** *Computing slow manifolds and canards by the continuation of orbit segments.* - Invited seminar talk, Numerical Analysis Seminar, Durham University, Durham (UK).

2013

- **25 January -** *Numerical continuation methods for slow-fast dynamical systems.* - Invited seminar talk, Séminaire: Géométrie hamiltonienne, Institut de Mathématiques de Jussieu, Université Pierre et Marie Curie, Paris (France).
- **20 May -** *Inflection methods in slow-fast dynamical systems.* - Invited talk in the mini-symposium Model reduction techniques for slow-fast systems, SIAM Conference on Application of Dynamical Systems, Snowbird (USA).
  Co-organiser of this mini-symposium with Morten Brøns.
- **4 June -** *Inflection lines near canards in 2D and 3D systems.* - Invited talk in the international conference "Slow-Fast Dynamics: Theory, Numerics and Applications to Life and Earth Sciences", Centre de Recerca Matemàtica, Barcelona (Spain).
  Co-organiser of this conference with Peter De Maesschalck, Mark A. Kramer, Martin Krupa, Serafim Rodrigues and Alexandre Vidal.
- **30 September -** *Inflection methods in slow-fast dynamical systems.* - Invited talk in the meeting Séminaire RNSC : Journées Couplées DyLeRBio/MnMs, INRA/INRIA, Montpellier (France).
  Co-organiser of this meeting with Fabien Campillo.

2012

- **29 March -** *Canards dans les systèmes dynamiques lents-rapides linéaires par morceaux du plan.* - Invited seminar talk, GT Dynamique Global des Systèmes Différentiels, Laboratoire Jacques-Louis Lions, Université Pierre et Marie Curie, Paris (France).
- **3 June -** *Numerical continuation techniques for planar slow-fast systems.* - Invited talk in the special session Singular Perturbations, The 9th AIMS Conference on Dynamical Systems, Differential Equations and Applications, Orlando (USA).
- **5 June -** *Canards, inflection and excitability threshold.* - Invited talk in the special session Multiple Time Scale Dynamics With a View Towards Biological Applications, The 9th AIMS Conference on Dynamical Systems, Differential Equations and Applications, Orlando (USA). Co-organiser of this mini-symposium with Martin Krupa and Alexandre Vidal.

- **13 September -** *Canards dans les systèmes dynamiques lents-rapides linéaires par morceaux du plan.* - Invited seminar talk, Séminaire du MAPMO, Département de Mathématiques, Université d'Orléans, Orléans (France).
- **20 September -** *Canards dans les systèmes dynamiques lents-rapides linéaires par morceaux du plan.* - Invited seminar talk, Séminaire BIBOP-CASYS, INRIA, Grenoble (France).
- **25 October -** *Méthodes de continuation numérique pour les systèmes dynamiques lents-rapides.* - Invited seminar talk, Séminaire du LMAH, Université du Havre, Le Havre (France).
- **3 December -** *Méthodes de continuation numérique pour les systèmes dynamiques lents-rapides; application aux dynamiques neuronales.* - Invited seminar talk, GdT Mathématiques et Neurosciences, Université Pierre et Marie Curie, Paris (France).
- **11 December -** *Numerical continuation methods for slow-fast dynamical systems – application to neuronal systems.* - Invited seminar talk, Séminaire: Modélisation et Calcul Scientifique, INRIA Paris-Rocquencourt, Le Chesnay (France).

2011

- **30 March -** *Numerical Continuation: Theory and Applications.* - Invited seminar talk, séminaire CalSim, Institut du Calcul et de la Simulation, Université Pierre et Marie Curie - Paris 6, Paris (France).
- **13 May -** *Méthodes de continuation numérique pour les perturbations singulières - application à certaines dynamiques neuronales..* - Invited seminar talk, séminaire MODEMIC, INRIA, Centre de Sophia-Antipolis, Montpellier (France).
- **22 May -** *Canard à l'orange: a new recipe for multiple time scales.* - Invited talk in the mini-symposium Multiple Time Scales Dynamics: from Theory to Biological Applications, SIAM Conference on Application of Dynamical Systems, Snowbird (USA).
  Co-organiser of this mini-symposium with Mike R. Jeffrey and Serafim Rodrigues.
- **29 June -** *Numerical Continuation: Theory, Implementation and Recent Advances.* - Invited seminar talk, Department of Computer Science and Artificial Intelligence, Universidad de Granada, Granada (Spain).
- **25 July -** *Numerical continuation of canard-like orbits in piecewise-linear systems.* - Invited talk in the mini-symposium Computational Methods, 7th European Nonlinear Oscillations Conference (ENOC), Rome (Italy).
- **26 July -** *Numerical continuation of isolas of equilibria.* - Invited talk in the mini-symposium Computational Methods, 7th European Nonlinear Oscillations Conference (ENOC), Rome (Italy).
- **17 August -** *Computation of invariant manifolds in slow-fast systems with mixed-mode oscillations..* - Invited seminar talk, Department of Mathematics, Nonlinear Analysis and Dynamics seminar, Technical University of Denmark, Copenhagen (Denmark).
- **15 September -** *Canards and Inflection.* - Invited presentation, Mathematical neuroendocrinology week, Regate research group, INRIA, Paris-Rocquencourt centre (France).

2010

- **7 January -** *Calcul numérique de variétés lentes dans les systèmes lents-rapides présentant des oscillations.* - Invited seminar talk, Séminaire d'Analyse et Applications, Université d'Évry Val d'Essonne, Evry (France).
- **15 February -** *Systèmes dynamiques lents-rapides en 2 et 3 dimensions: résultats théoriques et défis numériques.* - Invited seminar talk, Séminaire SISYPHE, INRIA, Centre de Paris-Rocquencourt (France).
- **17 February -** *Systèmes dynamiques lents-rapides en 2 et 3 dimensions: résultats théoriques et défis numériques.* - Invited seminar talk, Séminaire NeuroMathComp, INRIA, Centre de Sophia-Antipolis (France).
- **7 April -** *Computing 2D manifolds in slow-fast systems: a boundary value approach with an application to reaction dynamics.* - Invited talk in the special session *Dynamical Systems*, Joint British Mathematics Colloquium & British Applied Mathematics Colloquium, University of Edinburgh (UK).

- **4 June -** *Numerical continuation of isolas.* - Invited talk in the mini-symposium Recent Advances on Isolas and Invariant Manifolds: Numerics and Applications, SIAM Meeting DSPDEs'10, Barcelona (Spain).
  Co-organiser of this mini-symposium with Daniele Avitabile and Serafim Rodrigues
- **9 September -** *Canard à l'orange: a new recipe for multiple time scales.* - Invited talk in the mini-symposium Multiple Time Scale Behaviour, Dynamics Days Europe 2010, Bristol (UK).
  Co-organiser of this mini-symposium with Mike Jeffrey
- **4 October -** *Canards and inflection.* - Invited seminar talk, Department of Mathematics, Dynamical Systems Seminar, Boston University, Boston (USA).
- **7 December -** *Canards and inflection.* - Invited presentation, Workshop *Analyse Non-linéaire et Applications aux Rythmes du vivant (A.N.A.R.)*, Université de La Rochelle, La Rochelle (France).
- **13 December -** *Numerical Continuation in Slow-Fast Systems.* - Invited seminar talk, Department of Mathematics, Dynamical Systems Seminar, Hasselt University, Hasselt (Belgium).

2009

- **6 February -** *Numerical computation of slow manifolds and canards near a folded node.* - Invited seminar talk, Department of Mathematics, Dynamic Seminar, Cornell University, Ithaca (USA).
- **12 February -** *Numerical computation of slow manifolds and canards near a folded node.* - Invited seminar talk, Department of Mathematics, Cicada Seminar, The University of Manchester, Manchester (UK).
- **19 February -** *Detection and continuation of canard orbits.* - Invited presentation, Mini-Workshop on Slow-Fast Dynamics and Applications, University of Bristol, Bristol (UK).
- **20 March -** *Detection and continuation of canard orbits.* - Invited presentation, McGill University, Montréal (Canada).
- **23 March -** *Numerical computation of slow manifolds and canards near a folded node.* - Invited seminar talk, Department of Mathematics, Dynamical Systems Seminar, Boston University, Boston (USA).
- **16 April -** *Numerical continuation methods for slow-fast dynamical systems.* - Invited seminar talk, Department of Mathematics, Nonlinear dynamics and complex systems seminar, Technical University of Denmark, Copenhagen (Denmark).
- **20 May -** *Computation of global manifolds in slow-fast systems with mixed-mode oscillations.* - Invited talk in the mini-symposium Slow-fast Systems and Mixed-Mode Oscillations, SIAM Conference on Application of Dynamical Systems, Snowbird (USA).
  Co-organiser of this mini-symposium with Hinke M. Osinga.
- **21 October -** *Computation of global manifolds in slow-fast systems with mixed-mode oscillations.* - Invited seminar talk, School of Mathematical Sciences, Theoretical Mechanics Seminar, The University of Nottingham, Nottingham (UK).
- **2 December -** *Slow-fast systems and "canards" : from the theory to a numerical duck hunt!.* - Invited seminar talk, Department of Mathematics, Dynamics Seminar, University of Portsmouth, Portsmouth (UK).

2008

- **29 May -** *Calcul numérique de variétés lentes et de canards près d'une pseudo-singularité de type noeud.* - Invited seminar talk, Séminaire: Dynamique globale des systèmes différentiels, Université P. & M. Curie - Paris VI, France.
- **11 July -** *Slow manifolds, mixed-mode oscillations and canards in the self-coupled FitzHugh-Nagumo system.* - Invited talk in the mini-symposium Systems with Multiple Time Scales: Computations and Applications, 2008 SIAM annual meeting, San Diego (USA).
- **14 July -** *Numerical computation of slow manifolds and canards near a folded node.* - Invited seminar talk, Department of Mathematics, Mathematical Biology group, University of Pittsburgh, Pittsburgh (USA).

5.4 Contributed Talks and Posters

- **10 June 2015 -** *Slow-fast transitions in the Wendling-Chauvel neural mass model.* - 1st International Conference on Mathematical NeuroScience, Antibes - Juan les Pins, France.
- **15 May 2009 -** *Numerical computation of slow manifolds and canard orbits near a folded node - Application to neuronal dynamics.* - Deterministic and Stochastic Modeling in Computational Neuroscience and Other Biological Topics, Centre de Recerca Matemàtica, Barcelona, Spain.
- **31 March 2008 -** *Slow manifolds, mixed-mode oscillations and canards near a folded node.* - 50th British Applied Mathematics Colloquium, University of Manchester, UK.
- **23 April 2007 -** *The geometry of slow manifolds near a folded node.* - International Conference "Dynamics in Perturbations" on the occasion of Freddy Dumortier's 60th birthday, Hasselt University, Belgium.
- **18 April 2007 -** *Computing canards.* - 49th British Applied Mathematics Colloquium, University of Bristol, UK.

Poster *Computing slow manifolds and canards in the self-coupled FitzHugh-Nagumo system.*
Presented at :
ECMTB08, University of Edinburgh, UK **July 2008**.
Workshop in Nonlinear Neurodynamics, University of Exeter **September 2008.**

6 Grants and Awards

- **AXA grant** CENTAURE with Jean-Marc Lemaitre (Inserm, Montpellier) - 1.2M€ **2026-2030.**
(mécénat santé consortium with Inserm, Inria, CHU Toulouse)
- **ANR PRCI** AUDACITIES with Jorge Goncalves (Univ. of Luxembourg) - 800k€ **2026-2030.**
(consortium grant between Inria and U. of Luxembourg, in partnership with BCAM, Univ. Sherbrooke, Canada, and NIA, NIH, USA)
- **MIPS grant** SHEAFHD with Alexandre Bazin (LIRMM, Montpellier) - 80k€ **2026-2028.**
- **Inria Exploratory Action** 2MDS with Inria Astral (Bordeaux, France) and MCEN (BCAM, Bilbao, Spain) - 110k€ **2023-2025.**
- **Inria Associated Team** NeuroTransSF with MCEN (BCAM, Bilbao, Spain) - 25k€ **2019-2022.**
- **UCA collaborative grant** NeurFiMT - 20k€ **2018-2020.**
- **ANR seed grant** SloFaDyBio - 50k€ **2015-2017.**
- **RNSC grant** from the Complex Systems Institute of Paris-Île-de-France - 2k€ **2013.**
- **Postdoctoral Fellowship** awarded for three years by the Mathematical Biosciences Institute (MBI)
Offer declined since I accepted the permanent position at Inria. **February 2011.**
- **Visiting grant** "Research in Pair" awarded by the Centre International de Rencontres Mathématiques (C.I.R.M.) (Luminy, France) for a project together with Serafim Rodrigues **28 Feb - 12 Mar 2011.**
- **Postdoctoral Fellowship (Empower)** awarded for two years by the Irish Research Council for Science, Engineering & Technology (IRCSET) **July 2009.**
Offer decliner since I accepted the postdoctoral position at the University of Bristol.

- **Prize** awarded by the UK and Republic of Ireland section of SIAM together with the University of Manchester for **one of the best presentations by a PhD student at the British Applied Mathematics Colloquium 2008**, **April 2008**.
- **'Agrégation' Scholarship** given by the French Ministry of Education **2002-2003**.

7 Other academic activities

7.1 Editorial activities

- Co-Editor-in-Chief, SIAM book series on Mathematical Neuroscience, 2023-now
- Associate Editor, Discrete and Continuous Dynamical Systems - Series I (DCDS-I): Intelligence, 2025-now
- Co-editor of a special issue on "Excitable Dynamics in Neural and Cardiac Systems" of the journal Communications in Nonlinear Science and Numerical Simulation (IF 3.8), July 2020.

7.2 Conference and seminar organisation

- **Co-Chair of the International Conference on Mathematical Neuroscience 2027 (ICMNS '27)**, to be held in Montpellier (France), 1-4 June 2027.
- **Co-organizer of the EBRAINS Brain Simulation Workshop 2024**, held in Bilbao (Spain), 3-7 June 2024.
- **Co-organizer of the XXXXth of the Dynamics Days Europe Conference**, held at the University Côte d'Azur (Nice, France), 24-28 August 2021.
- **Co-organizer of the Dynamics Days Digital 2020 Online Conference**, 24-27 August 2020.
- **Program chair of the Waves Côte d'Azur Conference**, University of Nice (France), 4-7 June 2019.
- **Co-organizer of an International Conference** on
"Heteroclinic Dynamics in Neuroscience"
held at the J. A. Dieudonné Laboratory, University of Nice (France), 17-18 December 2015.
- **Co-organiser of an International Conference** on
"Multi-scale models, slow-fast differential equations, averaging in ecology"
held at the Centre Interfacultaire Bernoulli CIB, EPFL, Lausanne (Switzerland), 17-21 November 2014.
- **Co-organiser of an International Conference** on
"Slow-Fast Dynamics: Theory, Numerics and Applications to Life and Earth Sciences"
held at the Centre de Recerca Matemàtica, Barcelona (Spain), 3-7 June 2013.
- **Co-organiser of Mini-Symposia** at the following past international conferences
  - Coupled 80 Conference, Porto (Portugal), 09–10 October 2025.
  - SIAM Conference on Applications of Dynamical Systems, Denver (USA), 11–15 May 2025.
  - SIAM Conference on Applications of Dynamical Systems, Portland (USA), 14–18 May 2023.
  - Virtual Society for Mathematical Biology Annual Meeting, online, 13–17 June 2021.
  - SIAM Conference on Applications of Dynamical Systems, online, 23–27 May 2021.
  - XXXIXth Dynamics Days Europe Conference, Rostock (Germany), 2–6 September 2019.
  - SIAM Conference on Applications of Dynamical Systems, Snowbird (USA), 19–23 May 2019.
  - SIAM Conference on Applications of Dynamical Systems, Snowbird (USA), 21–25 May 2017.

- SIAM Conference on Applications of Dynamical Systems, Snowbird (USA), 17–21 May 2015.
- SIAM Conference on Applications of Dynamical Systems, Snowbird (USA), 19–23 May 2013.
- The 9th AIMS Conference on Dynamical Systems, Differential Equations and Applications, Orlando (USA), 1–6 June 2012.
- SIAM Conference on Applications of Dynamical Systems, Snowbird (USA), 22–26 May 2011.
- XXXth Dynamics Days Europe, Bristol, 6–10 September 2010.
- SIAM meeting Emerging Topics on Dynamical Systems and Partial Differential Equations, Barcelona (Spain), 31 May – 4 June 2010.
- SIAM Conference on Applications of Dynamical Systems, Snowbird (USA), 17–21 May 2009.

- **Organiser of the Applied Nonlinear Mathematics Seminar**, **University of Bristol**, Dept. of Eng. Mathematics **September 2009 - June 2011.**

7.3 Dynamical Systems community

Section chief editor, DSWeb Media Gallery **January 2010 - January 2016.**

7.4 Reviewing activities

Journals

- Automatica
- Bulletin of Mathematical Biology
- Biological Cybernetics
- Chaos: An Interdisciplinary Journal of Nonlinear Science
- Communications in Nonlinear Science and Numerical Simulations
- Discrete and Continuous Dynamical Systems - Series A
- Dynamical Systems: An International Journal
- Electronic Journal of Differential Equations
- Evolution Equations and Control Theory
- The IMA Journal of Applied Mathematics
- International Journal of Nonlinear Mechanics
- Journal de Mathématiques Pures et Appliquées
- Journal of Computational Neuroscience
- Journal of Mathematical Biology
- Journal of Nonlinear Science
- Molecular Biology of the Cell
- Neural Computation
- Nonlinear Differential Equations and Applications NoDEA
- Nonlinear Dynamics
- Nonlinearity
- Philosophical Transactions of the Royal Society, Series A
- Physica D
- Physics Letters A
- PLoS Computational Biology
- PNAS
- SIAM Journal on Applied Dynamical Systems
- SIAM Journal on Applied Mathematics
- SIAM Journal on Mathematical Analysis
- Scientific Reports

Funding bodies

- ANR (Agence Nationale de la Recherche, France)
- COST (EU)
- DFG, the German research foundation
- Leverhulme Trust (UK)
- FWO (Flemish Science Foundation, Belgium)
- Fondecyt (National Fund for Scientific and Technological Development, Chile)

7.5 Participation to Habilitation Thesis defences

- **Candidate:** Dr. Arnaud Tonnelier, Inria Rhône Alpes, Tripop Team, Grenoble, France.
  Date: 20 June 2019.
  Thesis: *Piecewise linear dynamical systems and excitability.*
  Role: Reviewer and jury member.

7.6 Participation to PhD Thesis defences

- **Candidate:** Abdelatif Oujbara, Le Havre Normandy University, Laboratory of Applied Mathematics of Le Havre (LMAH), Le Havre, France.
  Date: 1 October 2026.
  Thesis: *Numerical and Mathematical Analysis of Complex Dynamical Systems and Applications to Neuroscience.*
  Supervisors: Dr. Benjamin Ambrosio and Prof. Moulay Aziz-Alaoui.
  Role: Reviewer and Jury member.

- **Candidate:** Adel Malik Annabi, Inria Center at Université Côte d'Azur, project-team McTao, Sophia Antipolis, France.
  Date: 2 July 2026.
  Thesis: *State and Parameter Estimation of Neural Mass Models: A Control-Theoretic Approach.*
  Supervisors: Dr. Jean-Baptiste Pomet, Dr Dario Prandi and Dr. Ludovic Sacchelli.
  Role: Examiner.

- **Candidate:** Benjamin Böbel, Inria Center at Université Côte d'Azur, project-team Macbes, Sophia Antipolis, France.
  Date: 30 June 2026.
  Thesis: *Analysis of state and phase dynamics in a model of mammalian circadian clocks: synchronization and control properties.*
  Supervisors: Dr. Madalena Chaves and Dr. Jean-Luc Gouzé.
  Role: Examiner and President of the Jury.

- **Candidate:** Matteo Martin, Dept. of Information Engineering, University of Padova, Italy.
  Date: March 2026.
  Thesis: *Emergence of complex dynamics in neuronal and glial cells of the CNS.*
  Supervisors: Prof. Morten Gram Pedersen.
  Role: Reviewer.

- **Candidate:** Matthieu Aud'hui, LTSI, Inserm, Rennes, France.
  Date: 08 December 2025.
  Thesis: *L'épilepsie absence de l'enfant : comprendre et intervenir,.*
  Supervisors: Prof. Pascal Benquet.
  Role: Reviewer and President of the Jury.

- **Candidate:** Hélène Todd, Group for Neural Theory, ENS Paris, France.
  Date: 30 June 2025.
  Thesis: *Synchronous dynamics in networks of gap junction coupled inhibitory neurons,.*
  Supervisors: Dr. Boris Gutkin and Dr. N Alex Cayco Gajic.
  Role: Jury member.

- **Candidate:** Lisa Blum Moyse, Aistrosight Project-Team, Inria Lyon Centre, France.
  Date: 14 September 2023.
  Thesis: *Computational neuroscience models at different levels of abstraction for synaptic plasticity, astrocyte modulation of synchronization and systems memory consolidation.*

Supervisors: Dr. Hugues Berry.
Role: Reviewer and jury member.

- **Candidate:** Côme Le Breton, Athena Project-Team, Inria Center at Université Côte d'Azur, France.
  Date: 18 October 2022.
  Thesis: *Usability of phase synchrony neuromarkers in neurofeedback protocols for epileptic seizures reduction.*
  Supervisors: Dr. Théo Papadopoulo and Dr. Maureen Clerc.
  Role: Jury member and President of the Jury.

- **Candidate:** Evgenia Kartsaki, Biovision Project-Team, Inria Sophia Antipolis Research Centre, France.
  Date: 17 March 2022.
  Thesis: *How specific classes of retinal cells contribute to vision: a computational model.*
  Supervisors: Dr. Bruno Cessac and Prof. Evelyne Sernagor (University of Newcastle, UK).
  Role: Jury member.

- **Candidate:** Lou Zonca, Group of Computational Biology and Applied Mathematics, Institut de Biologie, École Normale Supérieure et PSL University, Paris, France.
  Date: 16 July 2021.
  Thesis: *Modeling and analytical computations of burst and interburst dynamics in neuronal networks, applications to neuron-glia interactions and oscillatory brain rhythms.*
  Supervisors: Prof. David Holcman.
  Role: Jury member.

- **Candidate:** Mattia Sensi, Dept. of Mathematics, University of Trento, Italy.
  Date: 18 Janvier 2021.
  Thesis: *A Geometric Singular Perturbation approach to epidemic compartmental models.*
  Supervisors: Prof. Andrea Pugliese.
  Role: Reviewer and jury member.

- **Candidate:** Marina Esteban, Dept. of Applied Mathematics, University of Seville, Spain.
  Date: 4 March 2019.
  Thesis: *Dynamics and Bifurcations in Nonlinear Systems with Hysteresis.*
  Supervisors: Prof. Enrique Ponce and Prof. Francisco Torres.
  Role: Reviewer and jury member.

- **Candidate:** Gabriela Capo Rangel, Basque Center for Applied Mathematics, Bilbao, Spain.
  Date: 12 February 2019.
  Thesis: *Computational predictive modeling of integrated cerebral metabolism, electrophysiology and hemodynamics.*
  Supervisors: Dr. Luca Gerardo-Giorda (BCAM) and Prof. Daniela Galvetti (Case Western Reserve University, USA).
  Role: Jury member.

- **Candidate:** Dora Karvouniari, Project-Biovision Team, Inria Sophia Antipolis Research Centre, France.
  Date: 15 March 2018.
  Thesis: *Mathematical modeling of retinal waves.*
  Supervisor: Dr. Bruno Cessac.
  Role: Reviewer and President of the Jury.

- **Candidate:** Catalina Vich Llompart, Dept. of Applied Mathematics, University of the Balearic Islands.
  Date: 5 July 2016.
  Thesis: *Inverse methods to estimate synaptic conductances with emphasis on non-smooth dynamical systems.*
  Supervisors: Prof. Antoni Guillamon and Prof. Rafel Prohens.
  Role: Examiner.

- **Candidate:** Elísabet Vela Felardo, Dept. of Applied Mathematics, University of Sevilla, Spain.
  Date: 18 July 2013.
  Thesis: *Piecewise Linear Differential Systems: Limit Cycles and Analysis of Bifurcations.*
  Supervisors: Prof. Enrique Ponce and Prof. Javier Ros.
  Role: Reviewer and jury member.

- **Candidate:** Soledad Fernández García, Dept. of Applied Mathematics, University of Sevilla, Spain.
  Date: 19 December 2012.
  Thesis: *Bifurcations of Periodic Orbits and Invariant Sets in Piecewise Linear Dynamical Systems.*
  Supervisors: Prof. Emilio Freire and Prof. Victoriano Carmona.
  Role: Reviewer and jury member.

8 Supervision activities

8.1 Mentoring of postdoctoral fellows

8.1.1 Present and planned for near future

- **Danillo Barros de Souza (MSCA fellow)**, **Inria Branch at the University of Montpellier**, MathNeuro project-team, France. **Project: Geometrical Data Analysis and AI study of Aging Longitudinal Datasets**. **Feb. 2027 - Jan. 2029.**

- **Sahil Imtiyaz (MIPS postdoc)**, **Inria Branch at the University of Montpellier**, MathNeuro project-team, France. **Project: Sheaf Theory for Biomedical Data**. **Nov. 2026 - March 2029.**

- **Rafael Cavalcanti (ANR postdoc)**, **Inria Branch at the University of Montpellier**, MathNeuro project-team, France. **Project: Topological Data Analysis in Aging**. **Nov. 2026 - Sept. 2028.**

8.1.2 Past

- Louisiane Lemaire, **Inria Branch at the University of Montpellier**, MathNeuro project-team, France. **Project: Multiscale modeling of Dravet syndrome**.
  [one joint article [77]] [Now postdoc at ENS Paris] **May. 2024 - April 2026.**

- Mattia Sensi, **Inria Sophia Antipolis Research Centre**, MathNeuro project-team, France.
  Project: Slow-fast dynamics of the neurotransmitter cycle.
  [one joint article [67]] [Now postdoc at Univ. Trento (Italy)] **Dec. 2021 - Mar. 2023.**

- Emre Baspinar, **Inria Sophia Antipolis Research Centre**, MathNeuro project-team, France.
  Project: Slow-fast dynamics of mean-field limits of stochastic neural networks.
  [one joint article [51]] [Now permanent researcher at Inria (Montpellier)] **Dec. 2018 - Apr. 2021.**

- Elif Köksal Ersöz, **Inria Sophia Antipolis Research Centre**, MathNeuro project-team, France.
  Project: Anticipated synchronisation in coupled slow-fast oscillators.
  [two joint articles: [39,44]][Now permanent researcher at Inria (Lyon)] **Jan. 2017 - Dec. 2018.**

- Soledad Fernández-García, **Inria Paris Research Centre**, Mycenae project-team, France.
  Project: piecewise-linear slow-fast dynamics, application to neuronal models.
  [three joint articles: [23,25,28], and one book chapter [Ch1]]
  [Now assistant professor at Univ. Sevilla (Spain)] **Dec. 2013 - Mar. 2016.**

- Albert Granados, **Inria Paris Research Centre**, Mycenae project-team, France.
  Project: slow-fast dynamics and averaging in neuronal models.
  [one joint article: [26]] [Now mathematics teacher in Barcelona (Spain)] **Dec. 2012 - Mar. 2015.**

8.2 PhD students (co-supervision)

8.2.1 Present and planned for near future

- **Ruizheng Guo**, **University College London (UCL) School of Pharmacy**, Ali Lab, London, UK. **"Physics Driven Computational Modelling of Synaptic Vesicle Dynamics in Alz.'s Disease"** [Co-supervision with Prof. A. Ali (UCL) and Prof S. Rodrigues (BCAM)] **Sept. 2026 - Sept. 2030**

- **Guillaume Darnatigues**, **Inria Branch at the University of Montpellier**, Inria Project-Team CAMIN and Inria Project-Team MathNeuro, Montpellier, France. **"Neural mass models to characterize the effect of stimulation on spinal cord neural networks."** **Nov. 2026 - Nov. 2030**
  [Co-supervision with Dr. T. Guiho (Inria, CAMIN)]

8.2.2 Past

- Guillaume Girier, **BCAM - Basque Center for Applied Mathematics**, Mathematical, Computational and Experimental Neuroscience (MCEN) research group, Bilbao, Spain. **"A Mathematical, Computational and Experimental study on neuronal excitability"** **Oct. 2020 - Sept. 2024**
  [three joint articles [66, 78, 79]] [Now postdoc at the Czech Acad. of Sci. (Prague)]

- Jordi Penalva Vadell, **University of the Balearic Islands**, Dept of Mathematics and Informatics, Palma, Spain. **"Mathematical study of slow-passage phenomena in piecewise-linear systems - Application to neuronal dynamics"** **Oct. 2020 - March 2024.**
  [two joint articles [63, 71]] [Now mathematics teacher in Palma (Spain)]

- Halgurd Taher, **Inria Sophia Antipolis Research Centre**, MathNeuro project-team, France. **"Mean-field limits of network of neural oscillators"** **Nov. 2018 - Dec. 2021**
  [one joint articles: [58]] [Now postdoc at Charité Berlin (Germany)]

- Louisiane Lemaire, **Inria Sophia Antipolis Research Centre**, MathNeuro project-team, France. **"Mathematical Modelling of Cortical Spreading Depression"** **Oct. 2018 - Dec. 2021.**
  [three joint articles: [54, 55, 65]] [Now postdoc at ENS Paris]

8.2.3 Participation in supervisions

- Elif Köksal Ersöz, **Inria Paris Research Centre**, Mycenae project-team, France. **"A mathematical study on coupled multiple timescale systems, synchronization of populations of endocrine neurons"** **Nov. 2013 - Nov. 2016.**
  [two joint articles: [27,34]] [Now permanent researcher at Inria (Lyon)]

- Giovanni Carmantini, **Plymouth University**, UK. **"Dynamical Systems Theory for Transparent Symbolic Computation in Neuronal Networks"**
  Defence date: 21 November 2016. **Oct. 2013 - Oct. 2016.**
  [one joint article and one joint proc. paper: [31, Proc6]] [Now Tech Entrepreneur (foldAI, CEO)]

- Lucile Megret, **Université Pierre et Marie Curie - Paris 6**, Laboratoire Jacques-Louis Lions, France. **"Explosion de cycles: analyse qualitative, simulations numériques et modèles"**
  Defence date: 25 November 2016. **Oct. 2013 - Oct. 2016.**
  [one joint proc. paper: [Proc5]] [Now lecturer at Sorbonne Université (Paris)]

8.3 Master and Bachelor students

- **Guillaume Darnatigues** (M2), **ENSTA**, France.
  "Modeling the spinal cord using neural mass models" **Spring / Summer 2026.**
- **Jan Gorecki** (BSc), **Wrocław University of Science and Technology**, Poland.
  "Synchronisation of bursting integrate and fire neurons with adaptation" **Summer 2026.**
- **Natalie Elena Cernei** (M2), **University of Bologna**, Italy.
  "Multiscale modeling of bursting activity in LHb neurons" **Fall 2025 / Winter 2026.**
- **Safaa Habib** (M2), **Université Côte d'Azur, Nice**, France.
  "Multiple-timescale analysis of a seizure model" **Summer-Fall 2023.**
- **Ufuk Cem Birbiri** (M1), **Université Côte d'Azur, Nice**, France.
  "Topological data analysis applied to aging" **Spring 2022.**
- **Mehmet Alihan Kayabas** (M2), **Sorbonne Université, Paris**, France.
  "Slow-fast dissection of a theta-gamma neural oscillator: single neuron and networks." **Spring 2022.**
- **Efstathios Pavlidis** (M2), **Université Côte d'Azur, Nice**, France.
  "Multiple-timescale dynamics, mixed mode oscillations and mixed affective states in a model of Bipolar Disorder." **Fall 2021 / Winter 2022.**
  [one joint article: [64]]
- **Efstathios Pavlidis** (M1), **Université Côte d'Azur, Nice**, France.
  "Analysis of a mathematical model of pituitary cell" **Spring 2021.**
- **Guillaume Girier** (M2), **Sorbonne Université, Paris**, France.
  "Revisiting resonator and integrator neuron models" **Spring 2020.**
- **Guillaume Girier** (M1), **Université Pierre et Marie Curie**, France.
  "Slow-fast dynamics in neural population models" **Summer 2019.**
- **Lucile Megret** (M2), **Université Pierre et Marie Curie**, Laboratoire Jacques-Louis Lions, France.
  "Dalí canards and spike-adding phenomenon in the Hindmarsh-Rose burster" **Winter 2013.**
- **Timothy Grundy** (M2), **University of Bristol**, Bristol Centre for Complexity Science, UK.
  "Canard-mediated synchronisation in coupled GnRH models" **Summer 2012.**

9 Teaching Activities

9.1 Master 2

Université de Montpellier, Sports Sci., Eng. and Ergonomics of Physical Activity, MSc Program, France.

Course (10 hrs, ca. 10 students) **2027-.**
Modeling in Neuroscience

9.2 Master 1

Université de Montpellier, Sports Sci., Eng. and Ergonomics of Physical Activity, MSc Program, France.

Course (10 hrs, ca. 10 students) **2025-.**
Modeling using Dynamical Systems

Sorbonne Université (Paris), Bio-Informatics Master Program, France.

Taught a Master Course (30 hrs, ca. 25 students) **Winters 2014-now**.
Mathematical and Computational Models in Neuroscience

Université Côte d'Azur, Modelling for neuronal and cognitive systems MSc Program, France.

Taught a Master Course (15 hrs, ca. 10 students) **2019 - 2025**.
Multiple-timescale dynamics in Neuroscience

University of Bristol, Department of Engineering Mathematics, UK.

Taught a 3rd-year course (30 hrs, ca. 20 students) **Winter 2012**.
Introduction to dynamical systems, complex variables and PDEs

Taught various tutorials, example classes and computer labs **2006-2009**.

- *Engineering Mathematics (tutorials)*
- *Mathematics with Maple (computer labs)*
- *Data Analysis with Matlab (computer labs)*
- *Discrete Mathematics (example classes)*